

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0166

Dell Power Station 138 kV
August 18, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0166. The Interconnection Customer proposes a 406.9 MW natural gas combined cycle facility interconnecting at the Dell Power Station 138 kV substation in the Virginia area. The request was received on October 20, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 112 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,962,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Centralia Substation / King City 138 kV, the Marlboro / King City 138 kV Line loads to 190.9% of its 168 MVA rating. Project DFAX: 65.9%.

For the loss of Los Molinos / Dell Power Station 138 kV line, the Dell Power Sta Substation / Grand Jct 138 kV loads to 186.5% of its 206 MVA rating. Project DFAX: 100.0%.

The Los Molinos / Dell Power Sta 138 kV is overloaded to 172.6% of its 223 MVA emergency rating for the loss of the Dell Power Station / Grand Jct 138 kV. The project contributes 100.0% DFAX (406.9 MW) to this overload.

The Harrogate / Grand Junction 138 kV is overloaded to 168.6% of its 198 MVA emergency rating for the loss of the Los Molinos / Dell Power Sta 138 kV. The project contributes 100.0% DFAX (406.9 MW) to this overload.

The Los Molinos / Fort Loudon Dam 138 kV is overloaded to 143.9% of its 241 MVA emergency rating for the loss of the Dell Power Station Substation / Grand Jct 138 kV. The project contributes 100.0% DFAX (406.9 MW) to this overload.

The King City / Centralia 138 kV is overloaded to 125.1% of its 245 MVA emergency rating for the loss of the Marlboro / King City 138 kV. The project contributes 60.9% DFAX (248.0 MW) to this overload.

Fort Loudon Dam / Richmond Energy 138 kV: 123.5% loading (257 MVA) following the outage of the Grand Jct / Dell Power Sta 138 kV; DFAX = 100.0%.

For the loss of King City / Centralia 138 kV, the Marlboro / Kevlar 138 kV loads to 112.8% of its 190 MVA rating. Project DFAX: 37.3%.

King City / Richmond Energy 138 kV: 112.1% loading (259 MVA) following the outage of the Grand Junction / Dell Power Station 138 kV L; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Marlboro / King City 138 kV	Rebuild line with higher-capacity conductor	\$2,196,000	100% (\$2,196,000)
Dell Power ta Substation / Grand Jct 138 kV	Rebuild line with higher-capacity conductor	\$3,702,000	100% (\$3,702,000)
Los Molinos / Dell Power Sta 138 kV	Rebuild line with higher-capacity conductor	\$2,028,000	100% (\$2,028,000)
Harrogate / Grand Junction 138 kV	Rebuild line with higher-capacity conductor	\$2,390,000	100% (\$2,390,000)
Fort Loudon Dam / Los Molinos 138 kV	Rebuild line with higher-capacity conductor	\$5,429,000	100% (\$5,429,000)
King City / Centralia 138 kV	Rebuild line with higher-capacity conductor	\$8,908,000	100% (\$8,908,000)
FtLoudon Dam / Richmond Energy 138 kV	Rebuild line with higher-capacity conductor	\$5,241,000	100% (\$5,241,000)
Marlboro / Kevlar 138 kV	Rebuild line with higher-capacity conductor	\$2,541,000	100% (\$2,541,000)
King City / Richmond Energy 138 kV	Rebuild line with higher-capacity conductor	\$2,527,000	100% (\$2,527,000)
	Total allocated to project		\$34,962,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.